

Report

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Introduction

This report analyzes the Haunted Places dataset, which documents paranormal events across the U.S., including location, descriptions, and sensory experiences. To explore possible external influences on these hauntings, we integrated five datasets—BRFSS Mental Health, Places of Worship, Alcohol Abuse, Time of Day, and Open Flights. This report presents key findings, clustering results, and insights into how this external information impacted hauntings.

BRFSS Mental Health Dataset

The BRFSS Mental Health dataset has MIME type text/csv. To extract mental health-related features, state-wide averages were calculated using the dataset attributes MENTHLTH and POORHLTH. MENTHLTH represents the number of days individuals reported poor mental health in the past 30 days, and POORHLTH indicates the number of days poor mental or physical health limited daily activities in the past 30 days. These attributes were used to derive “Average_Mental_Health_Days” and “Average_Poor_Health_Days,” respectively. The attribute ADDEPEV3 was used to calculate “Depression_Prevalence.” This column is a binary indicator of whether an individual has been diagnosed with a depressive disorder. Depression_Prevalence was computed as the proportion of “Yes” responses in ADDEPEV3 over the total number of responses. To ensure data consistency, special values were handled appropriately: responses coded as 88 (None) were treated as 0, reflecting no reported poor health days, while 77 (Don't know) and 99 (Refused) were set to NaN to prevent skewing the analysis. Additionally, any blank or missing responses were also assigned as NaN to maintain consistency in calculations.

Place of Worship Dataset

The Places of Worship dataset has a MIME type of application/geo+json. To extract geospatial features, both the Places of Worship and Haunted Places datasets were reprojected to a common coordinate reference system (EPSG:3395 - Mercator projection) to ensure accurate distance calculations. The feature "Distance_to_Nearest_Worship" was computed by determining the minimum distance from each haunted place to the closest place of worship. Using this, the "Haunted_Place_Proximity" feature was created as a Boolean flag to indicate whether a haunted place is within 5 miles (8,046.7 meters) of a place of worship. Lastly, the "Religion_Intersection" feature was assigned by identifying the religion of the nearest place of worship or marking it as "No Religion" if no place of worship was nearby.

Open Flights Dataset

For our analysis, we used the *Open Flights*¹ Dataset, an open source database containing 67663 routes spanning between 3321 airports worldwide. We joined this dataset with the *Our Airports*² dataset, another open source dataset containing descriptive and geolocational information for 82,562 airports worldwide. We filtered these datasets by only selecting active routes that start and end in the U.S. and airports that are open and are in the U.S. For each airport we interpolated a circular radius of influence that was sized according to airport type³. For each route we interpolated a Spherical Linear Interpolation.⁴

We added 4 direct features to our dataset: “Aerodrome_Counts”, “Aerodrome_Proximity”, “Flight_HighTraffic”, and “Flight_Intersection_Counts”. “Flight_Intersection_Counts” was calculated by counting the number of flights within 10 km of the haunted location. We followed a similar methodology for “Aerodrome_Counts”, assigning each airport a radius of influence based on its type and calculating intersections. “Flight_HighTraffic” was set to “True” if there were over 20 flight intersections and

¹ [Open Flights Dataset](#)

² [Our Airports Dataset](#)

³ [FAA Airspace Guidelines](#)

⁴ [SLERP](#)

“Aerodrome_Proximity” was set to “True” if a haunted place was within the radius of influence of any airport. Additionally, we stored detailed intersection data in two json dictionaries “airport_proximity_data.json” and “route_proximity_data.json” for plotting. All mathematical calculations were made using functions in “./dsci_550_a1/flightFunctions.py”. You can regenerate results and learn more about the methodology in notebook 2.01.

Although this data was downloaded as a .csv, we classified this dataset under the “Multipart” MIME type because we used the coordinates to produce an interactive html visualizations using the Plotly library in Python. We used these visualizations as an additional “feature” to investigate the haunted places dataset.

Daylight Duration Dataset

We utilized the [Sunrise-Sunset API](#) for precise daylight duration data, which offers advantages over TimeAndDate.com and U.S. Naval Observatory through batch processing capabilities and standardized JSON responses. We extracted “Daylight_Duration_Hours” and “Daylight_Category” features, providing more accurate results than alternatives that lack accessible APIs or rural location coverage. “Daylight_Duration_Hours” feature represents the precise number of daylight hours at each haunted location based on its latitude and longitude coordinates. “Daylight_Category” *locally* classified locations into three categories (Low, Medium, High) based on their daylight duration.

Time of Day and Daylight Results

Our analysis revealed significant patterns in how time and daylight affect hauntings. While investigating whether murders occur more frequently in the evening, we found no direct evidence of murder-specific clusters. This was likely due to the limited number of explicitly labeled murder events in the dataset and the tendency for violent events to be categorized under broader terms like “Violence” or “Supernatural | Violence” rather than specific murder classifications. Despite this limitation, we were able to analyze clusters containing violence-related events, which constitute approximately 11% of cases across the dataset. Death statistics showed 9-15% higher values during evening hours, with Jaccard cluster 2 showing evening deaths averaged 617.50 versus morning deaths at 463.00 (33.4% increase). Similarly, in Edit Distance cluster 7, evening deaths (615.00) exceeded both morning (588.50) and unknown times (532.91).

Apparition types varied distinctly by time: dusk showed highest Ghost concentrations (28.1%), mornings had more Spirit apparitions (8.0%), and evenings had highest Unknown types (71.6%). The majority (75.2%) of hauntings occur during 8-10 daylight hours, with peak activity at exactly 10 hours (48.2% of sightings). Different apparitions showed distinct daylight preferences: Spirit apparitions appeared more frequently in very short daylight periods (13.9%), more than double their overall dataset presence, while rarely appearing during long daylight (only 2.3%). Unknown apparitions were most prevalent during long daylight hours (76.3%), while Ghost apparitions preferred moderate daylight conditions. Evidence patterns reinforced time importance: visual evidence was highest at dusk (84.4%), while audio evidence was similar between evening (58.3%) and morning (57.7%) but noticeably lower during unknown times (44.9%). Both evidence types significantly depend on time of day ($p < 0.000001$).

Seasonally, 94.2% of haunting reports occur in winter ($p < 0.000001$), aligning with shorter daylight periods. The -0.3994 correlation between latitude and daylight hours suggests northern locations with shorter winter daylight create conditions conducive to paranormal experiences. These findings demonstrate that environmental factors significantly impact paranormal experiences, either by enhancing activity or affecting perception of ambiguous stimuli.

Places of Worship Dataset Results

The features extracted from the Places of Worship dataset were used to analyze the Haunted Places dataset. Clustering results using Jaccard similarity, edit distance, and cosine similarity revealed a strong correlation between proximity to religious sites and reported hauntings. Clusters with the highest number of haunted places were closest to worship centers. Jaccard similarity identified diverse apparition

types, with ghosts and spirits frequently reported near worship centers, especially in Cluster 2, which had the greatest religious diversity. Edit distance highlighted differences, showing that Clusters 3 and 4 had the most hauntings near Christian and Jewish sites, suggesting variations in how these traditions describe hauntings. Cosine similarity, however, formed a single large cluster, indicating shared commonalities across religious sites. Jaccard's effectiveness may stem from its binary nature, capturing the co-occurrence of apparition types and religious proximity without factoring in frequency differences, unlike cosine similarity, which focuses on vector direction. These findings suggest that religious beliefs influence how hauntings are perceived and reported. For example, Christian sites reported more ghosts and spirits, aligning with Christian views on purgatory, while Shinto and Pagan sites showed more spirit apparitions, consistent with ancestral spiritual beliefs. Overall, the analysis supports the hypothesis that religious settings shape the frequency and type of hauntings reported.

BRFSS Survey Results

The analysis of the CDC's 2023 BRFSS Survey found that higher mental health distress correlates with more ambiguous hauntings, particularly in apparition and event types. Jaccard similarity clustering showed that low-distress areas (Cluster 0) reported distinct apparitions, while high-distress areas (Clusters 1 and 2) had more unknown (62.5%) and violent hauntings (up to 17.86%), suggesting psychological distress may contribute to uncertain paranormal experiences. However, Jaccard was limited in capturing only the presence, not the frequency or distinctiveness, of apparition types. Edit distance, which formed five clusters, showed similar results, with Clusters 5 and 6, having the highest mental distress, reporting more unknown apparitions (42%) and spirits (14.6%), reinforcing that distress heightens sensitivity to vague supernatural experiences. Edit distance captured subtle variations in wording but didn't significantly alter overall trends. Cosine similarity, grouping all locations into one cluster, suggested that haunted places were similar in mental distress, apparition type, and event type, but oversimplified the data by treating all hauntings as a single category. These findings align with research on suggestibility, cognitive distortions, and emotional reactivity, indicating that stress and paranoia may lead to misinterpretations of vague haunted events.

Alcohol Dataset Results

Features from the Alcohol Abuse dataset, including Total Alcohol-Related Deaths, Underage Drinking Rates, and Time of Day, were analyzed to uncover insights about the Haunted Places dataset. The results suggest a strong correlation between alcohol abuse and hauntings, particularly in the evening and night. Jaccard similarity revealed that haunted places in high-alcohol states reported more ambiguous apparitions like Orbs, Spirits, and Shadow Figures. Cluster 2, with the highest alcohol-related deaths, showed 71.5% of hauntings at unknown times and 21.9% in the evening, suggesting impaired perception in low-light conditions leads to misidentifications. States with high alcohol-related deaths, such as California, Texas, Florida, and Pennsylvania, reported more hauntings, indicating alcohol-related cognitive distortions play a role in supernatural encounters. Southern and Western states, particularly those with high underage drinking rates, also showed more hauntings, hinting at regional cultural influences on supernatural perceptions. Edit distance clustering produced six clusters, with Clusters 5 and 6 (high alcohol-related deaths and underage drinking) showing the most reports of indistinct apparitions (42% Orbs, 14.6% Spirits), reinforcing that alcohol heightens sensitivity to unclear supernatural events. Cosine similarity, however, created a single cluster, suggesting haunted places were broadly similar in alcohol-related metrics, apparition type, and event type. Jaccard highlighted alcohol's correlation with nighttime hauntings, while edit distance showed how alcohol influences the perception and description of hauntings. The findings suggest that alcohol and darkness reduce sensory awareness, making people more likely to misinterpret ordinary occurrences as supernatural events, especially in high-alcohol states where unclear apparitions like Orbs and Spirits were more common.

Open Flights Results

In our feature extraction, we found 199 events relating to electronic malfunctions, plane crashes, and flying objects. We suspected that eye-witnesses could be mistaking overhead flights as “flying lights” and “orbs”. We also suspected that electronic malfunctions like “lights [...] [flickering] on & off”⁵ are caused by the high frequency electronics used by commercial aircraft. We hypothesize that these events would have elevated values for our added flight features. To confirm, we performed exploratory data analysis⁶ and clustering⁷ on these airborne related haunted places, the highlights of which are below.

We found that airborne-related events were below average in every flight related statistic. We believe this is because these haunted places are far from major transportation hubs which have a disproportionately high number of flights. We performed clustering on all flight related events to see if there were any commonalities between these entries. Edit distance clustering yielded the best separation, clustering together 2 plane crashes and 8 flying object sightings that were on average 1,000 meters away from a place of christian worship⁸. Additionally, we clustered a random sample of 1,000 haunted places using the added flight features; however, results were not informative. At its best, edit-distance made a cluster that had 3% of entries classified as “flying_objects”. Although this ratio is double that of the random samples, we attribute this to chance as no flight features were higher than the sample mean.

We also investigated what the haunted places could tell us about the Open Flights dataset. In particular, we wanted to know what flights we should avoid for Spring Break. Unsurprisingly LAX → LAS is in first place, intersecting with 738 haunted places. We also investigated airports finding that 3 out of the 10 most haunted heliports are in New Orleans. There are 6 haunted places within 3 miles of Disney's “Characters in Flight” balloon port, and the hill airforce base is the most haunted military base with 30 nearby haunted places.

We found that visualizing these features geographically was more insightful than clustering and summary statistics. Using the `plotly.py`⁹ library, we created an interactive map where you can select a subset of haunted places to plot and color code them by feature. We used the interpolated airport radii and flight routes to plot all intersecting routes and airports. You can explore the map in notebooks 3.01 and 3.02. Notebook 3.02 plots the most haunted airports mentioned above and Notebook 3.01 plots airborne related haunted places. Sample visualizations and observations can be found in the “Dans_Favs.md”.

Apparition Type Results

We analyzed apparition type, event type, and time of day to reveal patterns in hauntings and correlations between these features. Cosine similarity and Jaccard similarity produced similar clusters but differed in their approaches. Both methods identified Cluster 0 as the dominant cluster, which contained vague descriptions often marked as “Unknown,” due to missing details that make it difficult to categorize. Both methods also highlighted the influence of the time of day, with morning hauntings typically linked to ghosts and spirits. However, Cosine similarity grouped hauntings based on overall themes and time of day, leading to broader clusters, while Jaccard similarity focused on categorical overlaps.

However, in Cluster 3 Jaccard similarity grouped UFO and phantom sightings, despite their differences, due to similarities in their descriptions or time patterns, marking it as a rare mixed occurrence. Jaccard then created more distinct clusters, such as Cluster 2 (ghost-related) and Cluster 4 (spirit-related hauntings), which were reported mainly in the morning, suggesting time of day influences apparition types. For cosine similarity, Cluster 2 also revealed ghost sightings that also occurred in the morning, with more detailed descriptions focusing on the presence of ghosts. In contrast, Cluster 3 identified UFO activity, suggesting a multidimensional or hybrid nature of paranormal experiences in this

⁵ Index 1530

⁶ Notebook 4.01

⁷ */data/clustering/Flight_Features

⁸ Indices: [61, 926, 3739, 3961, 4111, 6033, 7137, 8722, 9408, 10408]

⁹ [plotly.py](#)

cluster. Lastly, Cluster 4 focused on spirit encounters, also primarily reported in the morning and linked to historical locations or traumatic events; indicating a different spiritual significance than a ghost sighting.

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Edit-Distance clustering identified more distinct clusters based on subtle event types and differences in apparitions. Cluster 2 (Flying Objects) and Cluster 3 (Phantoms) stood out due to their unique text differences, while Cluster 4 focused on violent hauntings, personal trauma, such as experiences of violence or death, could contribute to more intense or specific hauntings, often reported in the evening. The findings suggest that time of day played a significant role in categorizing hauntings, with morning hauntings linked to ghosts and spirits and evening hauntings associated with violent hauntings.

Apache Tika Thoughts

Apache Tika supports a wide variety of document formats and is effective at extracting metadata, making it a versatile tool for content analysis. Its documentation is helpful for troubleshooting and setting up the library. However, we encountered server errors when processing a large number of files, and processing times were slower with larger documents. Additionally, there were Python version compatibility issues during the installation of dependencies, which caused some delays. Despite these challenges, integrating Tika into Python via the REST API was straightforward, and once set up, it worked well for most document processing tasks. The main difficulties were managing large files and resolving installation issues.

Co-occurring Features and Locations

Our analysis identified recurring factors in haunted places. Areas with high mental distress showed more ambiguous hauntings, with Jaccard similarity linking depression prevalence to undefined apparition types. Proximity to places of worship also influenced hauntings, with clusters near religious sites reporting more ghost and spirit sightings. Jaccard similarity showed diverse apparition types near religious centers, while edit distance clustering highlighted descriptions near Christian and Jewish sites.

Alcohol abuse was another key factor, with high-alcohol areas reporting more ambiguous hauntings, particularly orbs and shadow figures. Jaccard similarity showed a strong correlation between alcohol-related deaths and nighttime hauntings, reinforcing the connection between impaired perception and paranormal interpretations.

Flight activity near haunted locations contributed to reports of electronic disturbances, flickering lights, and airborne anomalies. Edit distance clustering grouped haunted places near airports with descriptions of malfunctioning lights and unexplained flying objects. These findings indicate that hauntings are influenced by environmental, psychological, and external factors rather than supernatural events. Understanding these co-occurring features provides insight into how hauntings are perceived and reported.

Our daylight duration and time-of-day analysis provides further evidence that environmental factors significantly shape paranormal experiences. The strong correlation between specific light conditions and apparition types, combined with the seasonal concentration of hauntings during winter months with shorter daylight, suggests that natural light cycles create conditions that either enhance supernatural encounters or influence human perception of ambiguous stimuli.

Conclusion

Our analysis of the Haunted Places dataset, expanded with mental health, alcohol abuse, religious sites, and flight activity data, revealed key factors influencing paranormal reports. The findings suggest that hauntings are driven by individual perceptions, cultural narratives, and environmental conditions, such as mental distress, substance use, time of day, and proximity to religious or high-traffic areas, rather than supernatural causes. This study highlights how external factors shape paranormal experiences and provides opportunities for further research into the psychological and societal beliefs behind supernatural phenomena.